



Teacher Code : CHEY KERJ LOPF THOS WONJ TRIA RIMA FOXT

SPECIALIST MATHEMATICS UNIT 2

Tech-Free

Test theme: Complex Numbers

Reading Time: 5 minutes

Writing time: 45 minutes

QUESTION AND ANSWER BOOK

<i>Number of questions</i>	<i>Number of questions to be answered</i>	<i>Number of marks</i>
4	4	30

● Students are permitted to bring into the test room: pens, pencils, highlighters, erasers, sharpeners, rulers.

● Students are NOT permitted to bring into the examination: notes of any kind, blank sheets of paper, white out liquid/tape or a calculator of any type.

Materials supplied

- Question and answer book.
- Working space is provided throughout the book.

Instructions

- Write your **name** in the space provided above on this page.
- All written responses must be in English or Romanian.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the test room.

Instructions

Answer **all** questions in the spaces provided.

Unless otherwise specified an **exact** answer is required to a question.

In questions where more than one mark is available, appropriate working **must** be shown.

The city's main power line oscillates as a complex standing wave. To prevent the wires from melting, the 'Harmonic Displacement' z must satisfy a specific polynomial. The sensors indicate that the displacement z is a root of a polynomial where the coefficients are not real, suggesting a phase leak

Question 1 (7 marks)

The system governed by the equation:

$$z^3 + (2 - 2i)z^2 - 4iz - 8 = 0$$

- a. Show that $z = 2i$ is a solution to this equation

1 mark

- b. Use polynomial division or factorisation to find the other two roots in Cartesian form.

3 Marks

- c. Let the three roots be z_1, z_2, z_3 . These roots represent the 'Nodal Points' of the grid. If the 'Stability Index' is defined as the area of the triangle formed by these roots on the Argand diagram, calculate the Stability Index.

3 Marks

You move to the Sub-Level Inverter. Here, the current I is split into 'Prime Rotations'. The safety protocol requires you to find the exact values of the n -th roots of a complex power surge to synchronise the phases.

Question 2 (9 Marks)

A surge of $w = -8\sqrt{2} + 8\sqrt{2}i$ units has entered the inverter. To neutralise it, you must find the cube roots of w .

- a. Express w in the form $rcis(\theta)$, where $r > 0$ and $-\pi < \theta \leq \pi$.

2 Marks

- b. Determine the three cube roots of w , leaving them in polar form $rcis(\theta)$.

3 Marks

- c. Let z_k be the root with the smallest positive argument. To unlock the inverter, you must calculate the exact Cartesian form of z_k^4

4 Marks

The grid's heat shielding is failing. The 'Deflection Constant' D is calculated by summing the powers of a complex rotation. If D is not balanced, the shielding will shatter.

Question 3 (8 Marks)

Consider the complex number $u = \cos(\frac{\pi}{9}) + i\sin(\frac{\pi}{9})$. The Deflection Constant is given by:

$$D = \frac{u^{15}}{(\overline{u})^3}$$

- a. Simplify D into the form $x + iy$, giving exact values

3 Marks

- b. You are required to prove a safety identity for the shielding. Prove that:

$$(1 + \operatorname{cis}(\alpha))^n = (2\cos(\frac{\alpha}{2}))^n \operatorname{cis}(\frac{n\alpha}{2})$$

For any $\alpha \in (-\pi, \pi)$

3 Marks

- c. Using the identity from part b, find the exact cartesian form of $(1 + \operatorname{cis}(\frac{\pi}{3}))^6$

2 Marks

You reach the core. To save the city, you must solve the puzzle. The core's frequency is determined by the product of solutions to a cyclotomic equation.

Question 4 (6 Marks)

The core operates on the solutions to $z^5 + 1 = 0$.

- a. Solve $z^5 + 1 = 0$ for $z \in \mathbb{C}$. State your answers in polar form.

2 Marks

- b. Let the roots be w_0, w_1, w_2, w_3, w_4 . Show that $1 - w_0 + w_1 - w_2 + w_3 - w_4$ is not

the path to victory, but rather that the sum $\sum_{k=0}^4 w_k = 0$.

2 Marks

- c. The final code is the value of $\prod_{k=0}^4 (2 - w_k)$. Calculate the final Code.

2 Marks

THE GRID IS TURNING RED. BEGIN CALCULATIONS.

END OF TEST... UNTIL NEXT TIME...